Solution Design

Design goal

The goal of this POC is to prove that a focused AI-first support layer can reduce response time for routine inquiries, improve trust, and lower manual support load. I prioritized the smallest set of flows that can create measurable value quickly:

- product information
- order tracking
- returns and refunds
- human handoff

Architecture

The MVP uses a simple support architecture: a customer sends a message, the system detects intent, retrieves trusted business data where needed, and either resolves the request or escalates it to a human agent. This keeps the prototype easy to review while remaining production-friendly in shape

Core flow

The bot first classifies the request, then follows a grounded path for each supported intent. If key information is missing, it asks a clarifying question. If confidence is low, or if the case is sensitive, it hands off to a human agent. The returns/refunds flow is grounded in the supplied portfolio-company policy documents. See the Appendix for the full core flow diagram.

Key choices and prioritization

- `Product information` was prioritized because it supports pre-purchase conversion and can reduce cart abandonment.
- `Order tracking` was prioritized because it is one of the highest-volume repetitive support cases.
- `Returns/refunds` were prioritized because it affects post-purchase trust and satisfaction.
- `Human handoff` was included in the MVP because customer support systems should not guess sensitive or exception-heavy cases.

Technology stack

- runtime: Node.js
- server layer: native HTTP server
- UI: HTML, CSS, JavaScript web chat
- data layer: mocked product, order, and returns-policy data
- testing: Node built-in test runner
- optional AI layer: OpenAI Responses API for grounded reply composition when credentials are present

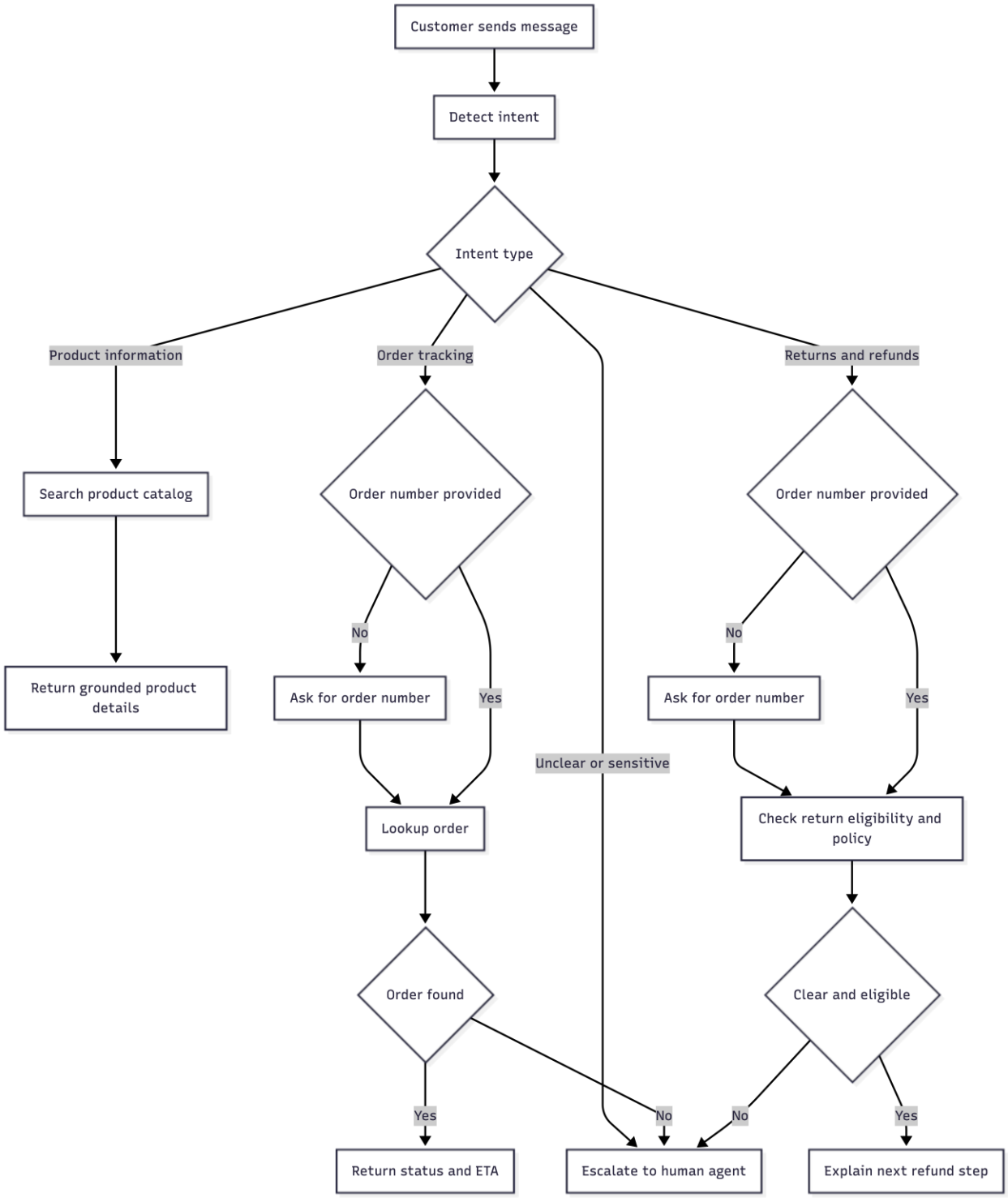
Integration potential

The design can connect cleanly to common commerce and support systems:

- product catalog: Shopify, Medusa, or a custom commerce backend
- order service: OMS, ERP, or courier tracking providers
- handoff/CRM: Zendesk, Freshdesk, Intercom, HubSpot, or internal tooling
- analytics: BI or product analytics systems for response time, resolution rate, and support deflection

Guardrails and scalability

- factual answers are grounded in structured business data
- missing information triggers clarifying questions
- low-confidence or sensitive cases escalate to human support
- the architecture is bilingual-ready and can expand to new languages in the future

Appendix

Appendix A. Core Dialog Flow Diagram